LOURDES MATHA

COLLEGE OF SCIENCE AND TECHNOLOGY

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MOBILE APPLICATION DEVELOPMENT LAB (20MCA243)

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From Page No: .....To:…….

Certified Bonafide Record of Work Done By

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Date: 08/07/26

EXPERIMENT 1

Aim: Design a Login page with username and password using Linear Layout and Toast Valid and Invalid credentials.

ALGORITHM

1. Start the application and display the login layout.
2. Initialize the username field, password field and Login button.
3. Wait for the user to enter a username and password and tap Login.
4. Read both input values as strings.
5. Compare the username with usr and the password with 123.
6. If both values match, display a login success Toast; otherwise, display an invalid credentials Toast.
7. Allow the user to try again or close the application.

Program Code:

MainActivity.java

```
package com.example.myapplication;
import androidx.appcompat.app.AppCompatActivity;

import android.os.Bundle;
import android.view.View;
import android.widget.Button;
import android.widget.EditText;
import android.widget.Toast;

public class MainActivity extends AppCompatActivity {
   Button b;
   EditText u,p;
   @Override
   protected void onCreate(Bundle savedInstanceState) {
      super.onCreate(savedInstanceState);
      setContentView(R.layout.activity_main);
      b=(Button) findViewById(R.id.b);
      u=(EditText) findViewById(R.id.user);
      p=(EditText) findViewById(R.id.pass);
      b.setOnClickListener(new View.OnClickListener() {
         @Override
         public void onClick(View view) {
            String user=u.getText().toString();
            String pass=p.getText().toString();
            if (user.equals("usr")&&  pass.equals("123"))
```

```
            Toast.makeText(getApplicationContext(),"logged in
successfully",Toast.LENGTH_SHORT).show();
          else
            Toast.makeText(getApplicationContext(),"Invalid
credentials",Toast.LENGTH_SHORT).show();
        }})
  }
}
```

Activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout
xmlns:android="http://schemas.android.com/apk/re
s/android"
  xmlns:app="http://schemas.android.com/apk/res-
auto"
  xmlns:tools="http://schemas.android.com/tools"
  android:layout_width="match_parent"
  android:layout_height="match_parent"
  tools:context=".MainActivity"
android:background="@drawable/ic_launcher_bac
kground">
  <RelativeLayout
    android:layout_width="match_parent"
    android:layout_height="match_parent">

    <TextView
      android:id="@+id/l"
      android:layout_width="1200dp"
      android:layout_height="wrap_content"
      android:layout_alignParentEnd="true"
      android:layout_marginTop="25dp"
      android:layout_marginEnd="0dp"
      android:background="#B23B1E40"
      android:text="LOGIN"
      android:textAlignment="center"
      android:textColor="#ffffff"
      android:textSize="30dp" />

    <TextView
      android:id="@+id/u"
      android:layout_width="wrap_content"
      android:layout_height="wrap_content"
      android:text="Username:"
      android:layout_marginTop="100dp"
      android:layout_marginLeft="20dp"
      android:textSize="20dp”
      />

<TextView
```

```
        android:id="@+id/textView3"
        android:layout_width="wrap_content"
```
Result: The program has executed successfully and output is verified

```
        android:layout_height="wrap_content"
        android:text="Password:"
        android:layout_marginTop="150dp"
        android:layout_marginLeft="20dp"
        android:textSize="20dp"
        />
<Button
        android:id="@+id/b"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:layout_marginLeft="150dp"
        android:layout_marginTop="250dp"
android:backgroundTint="#DF803BBC"
        android:text="login"
        android:textSize="20dp" />

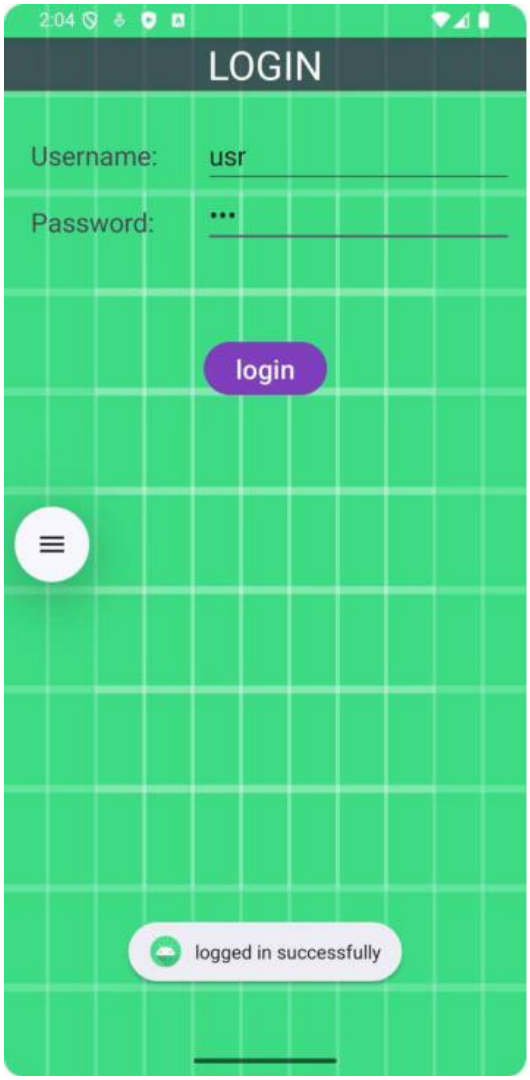
      <EditText
        android:id="@+id/user"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:layout_marginTop="90dp"
        android:layout_marginLeft="150dp"
        android:textSize="20dp"
        android:ems="10"
        android:inputType="text"
        android:text="" />

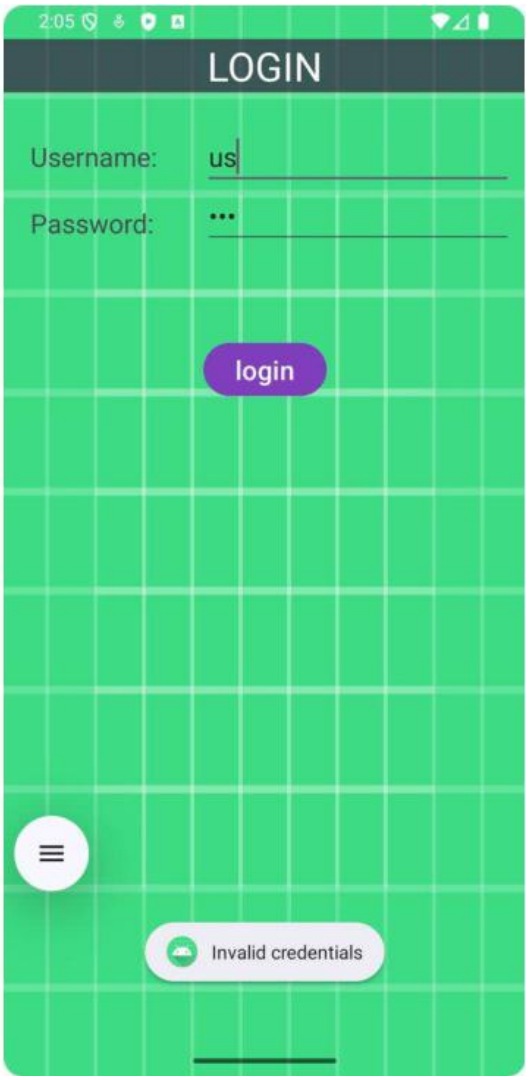
      <EditText
        android:id="@+id/pass"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:layout_marginTop="135dp"
        android:layout_marginLeft="150dp"
        android:textSize="20dp"
        android:ems="10"
        android:inputType="textPassword" />
  </RelativeLayout>

</LinearLayout>
```

Result: The program has executed successfully and output is verified

OUTPUT


2:04
LOGIN
Username:
usr
Password:
login
logged in successfully


2:05
LOGIN
Username:
us
Password:
login
Invalid credentials

Date : 14/07/26

EXPERIMENT 2

Aim: Implement Explicit Intent for Login (1st activity contains login details, when clicked to submit, go to 2nd activity)

ALGORITHM

1. Start the application and display the login activity.
2. Initialize the username field, password field and Login button.
3. Read the entered username and password when Login is tapped.
4. Compare the entered values with the credentials specified in the program.
5. For valid credentials, show a success Toast and create an explicit Intent for the Homepage activity.
6. Start the Homepage activity and display the welcome message and Logout button.
7. For invalid credentials, remain on the login activity and show an error Toast.
8. When Logout is tapped, start MainActivity using an explicit Intent and finish the Homepage activity.

Program Code:

MainActivity.java

```
package com.example.login;
import android.content.Intent;
import android.os.Bundle;
import android.view.View;
import android.widget.Button;
import android.widget.EditText;
import android.widget.Toast;
import androidx.activity.EdgeToEdge;
import androidx.appcompat.app.AppCompatActivity;
import androidx.core.graphics.Insets;
import androidx.core.view.ViewCompat;
import androidx.core.view.WindowInsetsCompat;

public class MainActivity extends AppCompatActivity {
   EditText e1,e2;
   Button b1;
   @Override
   protected void onCreate(Bundle savedInstanceState) {
      super.onCreate(savedInstanceState);
      EdgeToEdge.enable(this);
      setContentView(R.layout.activity_main);
      ViewCompat.setOnApplyWindowInsetsListener(findViewById(R.id.main), (v, insets) -> {
         Insets systemBars = insets.getInsets(WindowInsetsCompat.Type.systemBars());
         v.setPadding(systemBars.left, systemBars.top, systemBars.right, systemBars.bottom);
         return insets;
```

```
    });
    e1=(EditText) findViewById(R.id.edittext1);
    e2=(EditText) findViewById(R.id.edittext2);
    b1=(Button) findViewById(R.id.btn1);
    b1.setOnClickListener(new View.OnClickListener() {
      @Override
      public void onClick(View v) {
        String username=e1.getText().toString().trim();
        String pass=e2.getText().toString();
        if(username.equals("user")  && pass.equals("admin123")) {
          Toast.makeText(MainActivity.this, "Login successful", Toast.LENGTH_SHORT).show();
          Intent intent = new Intent(MainActivity.this, Homepage.class);
          startActivity(intent);
        }
        else
          Toast.makeText(MainActivity.this,"Login  failed",Toast.LENGTH_SHORT).show();
      }
    });
  }}
```

Activity_Main.xml

```
<androidx.constraintlayout.widget.ConstraintLay
out
xmlns:android="http://schemas.android.com/apk/r
es/android"
xmlns:app="http://schemas.android.com/apk/res-
auto"
   android:id="@+id/main"
   android:layout_width="match_parent"
   android:layout_height="match_parent">

   <TextView
      android:id="@+id/textView3"
      android:layout_width="wrap_content"
      android:layout_height="wrap_content"
      android:text="Login"
      android:textSize="24sp"
      android:textStyle="bold"
      app:layout_constraintTop_toTopOf="parent"
     app:layout_constraintStart_toStartOf="parent"
      app:layout_constraintEnd_toEndOf="parent"
      android:layout_marginTop="64dp" />

   <TextView
      android:id="@+id/textView4"
      android:layout_width="wrap_content"
      android:layout_height="wrap_content"
      android:text="Username"
app:layout_constraintTop_toBottomOf="@id/text
```

```
View3"
     app:layout_constraintStart_toStartOf="parent"
       android:layout_marginTop="32dp"
       android:layout_marginStart="32dp" />
   <EditText
       android:id="@+id/edittext1"
       android:layout_width="0dp"
       android:layout_height="wrap_content"
       android:hint="Enter username"
       android:inputType="text"
app:layout_constraintStart_toEndOf="@id/textVi
ew4"
app:layout_constraintEnd_toEndOf="parent"
app:layout_constraintTop_toTopOf="@id/textVie
w4"
       android:layout_marginStart="16dp"
       android:layout_marginEnd="32dp" />
   <TextView
       android:id="@+id/textView5"
       android:layout_width="wrap_content"
       android:layout_height="wrap_content"
       android:text="Password"
app:layout_constraintTop_toBottomOf="@id/edit
text1"
app:layout_constraintStart_toStartOf="parent"
       android:layout_marginTop="24dp"
       android:layout_marginStart="32dp" />

   <EditText
       android:id="@+id/edittext2"
       android:layout_width="0dp"
       android:layout_height="wrap_content"
       android:hint="Enter password"
       android:inputType="textPassword"
app:layout_constraintStart_toEndOf="@id/textVi
ew5"
app:layout_constraintEnd_toEndOf="parent"
app:layout_constraintTop_toTopOf="@id/textVie
w5"
       android:layout_marginStart="16dp"
       android:layout_marginEnd="32dp" />

   <Button
       android:id="@+id/btn1"
       android:layout_width="wrap_content"
```

Homepage.java

```
package com.example.login;
import android.content.Intent;
import android.os.Bundle;
```

```
import android.view.View;
import android.widget.Button;
import androidx.activity.EdgeToEdge;
import androidx.appcompat.app.AppCompatActivity;
import androidx.core.graphics.Insets;
import androidx.core.view.ViewCompat;
import androidx.core.view.WindowInsetsCompat;

public class Homepage extends AppCompatActivity {
   Button logoutButton;
   @Override
   protected void onCreate(Bundle savedInstanceState) {
      super.onCreate(savedInstanceState);
      EdgeToEdge.enable(this);
      setContentView(R.layout.activity_homepage);
      ViewCompat.setOnApplyWindowInsetsListener(findViewById(R.id.main), (v, insets) -> {
         Insets systemBars = insets.getInsets(WindowInsetsCompat.Type.systemBars());
         v.setPadding(systemBars.left, systemBars.top, systemBars.right, systemBars.bottom);
         return insets;
      });

      logoutButton = findViewById(R.id.logoutButton);
      logoutButton.setOnClickListener(new View.OnClickListener() {
         @Override
     public void onClick(View v) {
           // Go back to login screen
      android:layout_height="wrap_content"
      android:text="Login"
app:layout_constraintTop_toBottomOf="@id/edit
text2"
app:layout_constraintStart_toStartOf="parent"
      app:layout_constraintEnd_toEndOf="parent"
      android:layout_marginTop="32dp" />
</androidx.constraintlayout.widget.ConstraintLay
out>
            Intent intent = new Intent(Homepage.this, MainActivity.class);
            startActivity(intent);
            finish();
         }
      });
   }}
```

Activity_HomePage.xml

```
<?xml version="1.0" encoding="utf-8"?>
<androidx.constraintlayout.widget.ConstraintL
ayout
xmlns:android="http://schemas.android.com/ap
k/res/android"
xmlns:app="http://schemas.android.com/apk/re
s-auto"
   android:layout_width="match_parent"
```

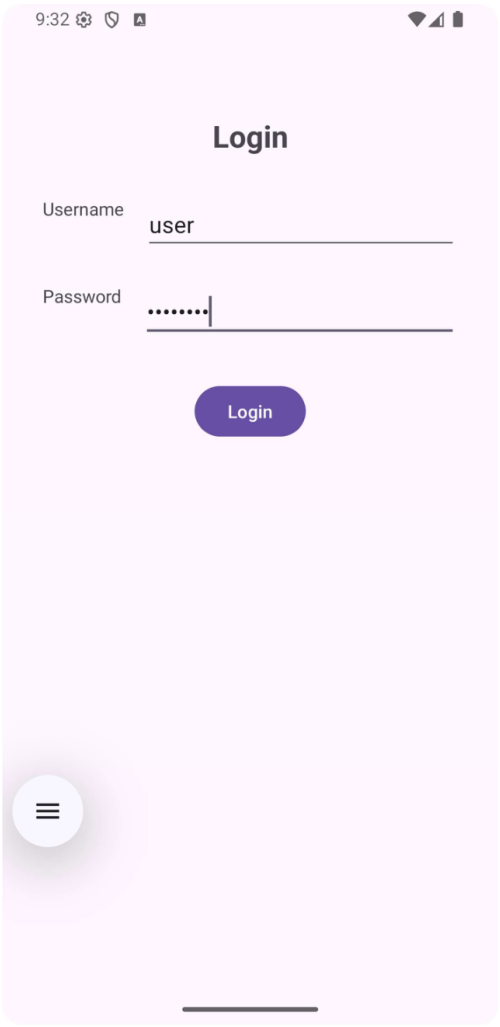
```
    android:layout_height="match_parent"
    android:id="@+id/main"
    android:padding="24dp">

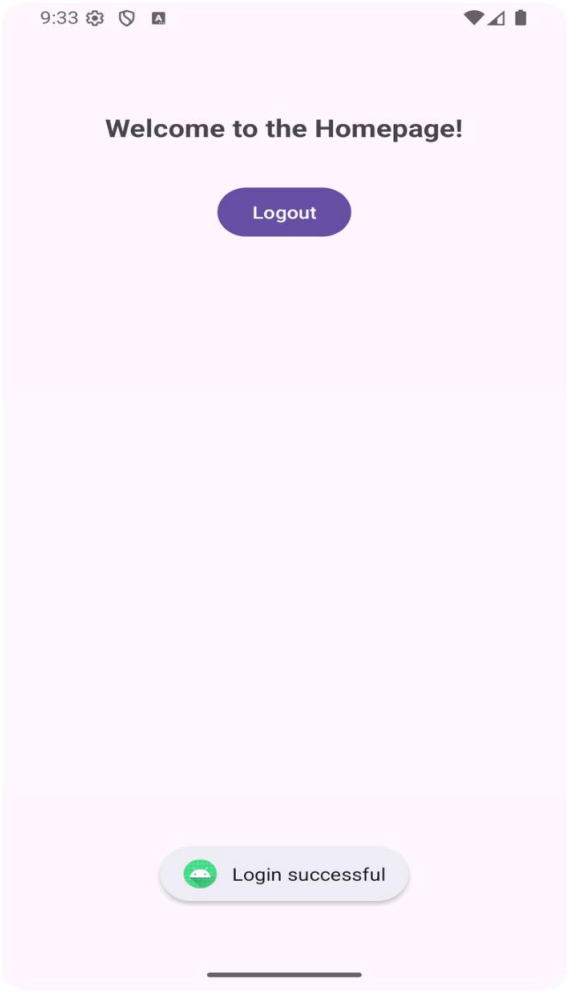
    <TextView
        android:id="@+id/welcomeText"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Welcome to the
Homepage!"
        android:textSize="20sp"
        android:textStyle="bold"
app:layout_constraintTop_toTopOf="parent"
app:layout_constraintStart_toStartOf="parent"
app:layout_constraintEnd_toEndOf="parent"
        android:layout_marginTop="64dp" />
    <Button
        android:id="@+id/logoutButton"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Logout"
Result: The program has executed successfully and output is verified
app:layout_constraintTop_toBottomOf="@id/
welcomeText"

app:layout_constraintStart_toStartOf="parent"
app:layout_constraintEnd_toEndOf="parent"
        android:layout_marginTop="32dp" />
</androidx.constraintlayout.widget.Constraint
Layout>
```

Result: The program has executed successfully and output is verified

OUTPUT


9:32
Login
Username
user
Password
Login


9:33
Welcome to the Homepage!
Logout
Login successful

Date : 19/07/26

EXPERIMENT 3

Aim: Create a simple calculator to perform arithmetic operations using Grid Layout.

ALGORITHM

1. Start the application and display the calculator layout.

2. Initialize the display, digit buttons, operator buttons, decimal button, delete button, equals button and AC button.

3. Append each tapped digit to the current display value; allow a decimal point only when one is not already present.

4. When an arithmetic operator is tapped, store the first number and operator, then clear the display for the second number.

5. Read the second number when the equals button is tapped.

6. Perform addition, subtraction, multiplication or division according to the stored operator.

7. Display the result and reset the stored first number and operator.

8. Use delete to remove the last character or AC to clear the display; continue accepting input until the application is closed.

Program Code:

MainActivity.java

```
package com.example.calculator;
import androidx.appcompat.app.AppCompatActivity;
import android.os.Bundle;
import android.view.View;
import android.widget.Button;
import android.widget.EditText;
public class MainActivity extends AppCompatActivity
{
   EditText t1;
   Button bac;
   String firstNumber="",operator="";
   @Override
   protected void onCreate(Bundle savedInstanceState)
   {
      super.onCreate(savedInstanceState);
      setContentView(R.layout.activity_main);
      t1=(EditText) findViewById(R.id.t1);
      bac=(Button) findViewById(R.id.bac);
      int b[]=new int[17];
      for(int i=0;i<=16;i++)
```

```
    {
        String s="b"+(i);
        int rid=getResources().getIdentifier(s,"id",getPackageName());
        b[i]=rid;
    }
    for(int id:b)
    {
        Button button=findViewById(id);
        if(button !=null)
        {
            button.setOnClickListener(new View.OnClickListener()
            {
                @Override
                public void onClick(View view)

                {
                    handleButtonClick(view.getId());
                }});
    }}

    bac.setOnClickListener(new View.OnClickListener()
    {
        @Override
        public void onClick(View view)
        {
            t1.setText("");
        }});
}

private void handleButtonClick(int id)
{
    String txt = t1.getText().toString();
    if (id == R.id.b16)
    {
        if (!firstNumber.isEmpty() && !operator.isEmpty())
        {
            String secondNumber = txt;
            double result = 0;
            double num1 = Double.parseDouble(firstNumber);
            double num2 = Double.parseDouble(secondNumber);

            switch (operator)
            {
                case "+":
                    result = num1 + num2;
                    break;
                case "-":
                    result = num1 - num2;
                    break;
                case "x":
```

```
                result = num1 * num2;
                break;
            case "/":
                result = num1 / num2;
                break;
        }
        t1.setText(String.valueOf(result));
        firstNumber = "";
        operator = "";
      }}
    else if (id == R.id.b10)
      if (!txt.contains("."))
t1.setText(txt + ((Button) findViewById(id)).getText());
    else if (id == R.id.b12 || id == R.id.b13 || id == R.id.b14 || id == R.id.b15)
      if (!txt.isEmpty())
      {
        firstNumber = txt;
        operator = ((Button) findViewById(id)).getText().toString();
        t1.setText("");
      }
    else if (id == R.id.b11)
      if (txt.length() > 0)
        t1.setText(txt.substring(0, txt.length() - 1));
    else
      t1.setText(txt + ((Button) findViewById(id)).getText());
  }}
```

Activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout
xmlns:android="http://schemas.android.com/ap
k/res/android"
xmlns:app="http://schemas.android.com/apk/re
s-auto"
xmlns:tools="http://schemas.android.com/tools
"   android:layout_width="match_parent"
   android:layout_height="match_parent"
   android:background="#000000"
   tools:context=".MainActivity">

   <GridLayout
      android:layout_width="match_parent"
      android:layout_height="match_parent"
      android:columnCount="4"
      android:rowCount="6"
      android:layout_margin="5dp">

      <EditText
        android:id="@+id/t1"
        android:layout_marginTop="350dp"
```

```
        android:layout_marginBottom="15dp"
        android:layout_width="match_parent"
        android:layout_height="150dp"
        android:layout_row="0"
        android:layout_column="0"
        android:ems="10"
        android:layout_columnSpan="4"
        android:inputType="text"
        android:textSize="92px"
        android:textAlignment="viewEnd"
        android:background="#20ffffff"
        android:textColor="#ffffff"
        android:text=""/>
android:background="@drawable/button_back
ground"
        android:gravity="center"
        android:text="2"
        android:textAlignment="center"
        android:textSize="22dp"
        android:textStyle="bold"
       android:layout_marginBottom="10dp"/>

    <Button
        android:id="@+id/b3"
        android:layout_width="105dp"
        android:layout_height="75dp"
        android:layout_row="1"
        android:layout_column="2"
        android:layout_marginLeft="5dp"
android:background="@drawable/button_back
ground"
        android:text="3"
        android:gravity="center"
        android:textAlignment="center"
        android:textSize="28dp"
        android:textStyle="bold"
       android:layout_marginBottom="10dp"/>

    <Button
        android:id="@+id/b12"
        android:layout_width="105dp"
        android:layout_height="75dp"
        android:layout_marginLeft="5dp"
        android:layout_row="1"
        android:layout_column="3"
        android:gravity="center"
        android:textAlignment="center"
        android:textSize="28dp"
        android:textStyle="bold"
        android:layout_marginBottom="10dp"
```

```
        android:background="@drawable/op_bg"
          android:textColor="#ffffff"
          android:text="+"/>

      <Button
          android:id="@+id/b4"
          android:layout_width="105dp"
          android:layout_height="75dp"
          android:layout_marginLeft="5dp"
          android:layout_row="2"
          android:layout_column="0"
          android:gravity="center"
          android:textAlignment="center"
          android:textSize="28dp"
          android:textStyle="bold"
          android:layout_marginBottom="10dp"
android:background="@drawable/button_back
ground"  android:text="4"/>

      <Button
          android:id="@+id/b5"
          android:layout_width="105dp"
          android:layout_height="75dp"
          android:layout_marginLeft="5dp"
          android:layout_row="2"
          android:layout_column="1"
          android:gravity="center"
          android:textAlignment="center"
          android:textSize="28dp"
          android:textStyle="bold"
          android:layout_marginBottom="10dp"
android:background="@drawable/button_back
ground" android:text="5"/>

      <Button
          android:id="@+id/b6"
          android:layout_width="105dp"
          android:layout_height="75dp"
          android:layout_marginLeft="5dp"
          android:layout_row="2"
          android:layout_column="2"
android:background="@drawable/button_back
ground" android:gravity="center"
          android:textAlignment="center"
          android:textSize="28dp"
          android:textStyle="bold"
          android:layout_marginBottom="10dp"
          android:text="6"/>

      <Button
```

```
        android:id="@+id/b13"
        android:layout_width="105dp"
        android:layout_height="75dp"
        android:layout_marginLeft="5dp
android:layout_row="2"
        android:layout_column="3"
android:background="@drawable/op_bg"
        android:textColor="#ffffff"
android:gravity="center"
        android:textAlignment="center"
        android:textSize="28dp"
        android:textStyle="bold"
        android:layout_marginBottom="10dp"
        android:text="-"/>

    <Button
        android:id="@+id/b7"
        android:layout_width="105dp"
        android:layout_height="75dp"
        android:layout_marginLeft="5dp"
        android:layout_row="3"
        android:layout_column="0"
        android:gravity="center"
        android:textAlignment="center"
        android:textSize="28dp"
        android:textStyle="bold"
        android:layout_marginBottom="10dp"
android:background="@drawable/button_back
ground" android:text="7"/>

    <Button
        android:id="@+id/b8"
        android:layout_width="105dp"
        android:layout_height="75dp"
        android:layout_marginLeft="5dp"
        android:layout_row="3"
        android:layout_column="1"
        android:gravity="center"
        android:textAlignment="center"
        android:textSize="28dp"
        android:textStyle="bold"
        android:layout_marginBottom="10dp"
android:background="@drawable/button_back
ground" android:text="8"/>

    <Button
        android:id="@+id/b9"
        android:layout_width="105dp"
        android:layout_height="75dp"
        android:layout_marginLeft="5dp"
```

```
        android:layout_row="3"
        android:layout_column="2"
        android:gravity="center"
        android:textAlignment="center"
        android:textSize="28dp"
        android:textStyle="bold”
android:layout_marginBottom="10dp"
android:background="@drawable/button_back
ground" android:text="9"/>

    <Button
        android:id="@+id/b14"
        android:layout_width="105dp"
        android:layout_height="75dp"
        android:layout_marginLeft="5dp"
        android:layout_row="3"
        android:layout_column="3"
        android:gravity="center"
        android:textAlignment="center"
        android:textSize="28dp"
        android:textStyle="bold"
        android:layout_marginBottom="10dp"
android:background="@drawable/op_bg"
        android:textColor="#ffffff"
        android:text="x"/>

    <Button
        android:id="@+id/b0"
        android:layout_width="105dp"
        android:layout_height="75dp"
        android:layout_marginLeft="5dp"
        android:layout_row="4"
        android:layout_column="0"
        android:gravity="center"
        android:textAlignment="center"
        android:textSize="28dp"
        android:textStyle="bold"
        android:layout_marginBottom="10dp"
android:background="@drawable/button_back
ground" android:text="0"/>
    <Button
        android:id="@+id/b10"
        android:layout_width="105dp"
        android:layout_height="75dp"
        android:layout_marginLeft="5dp"
        android:layout_row="4"
        android:layout_column="1"
        android:gravity="center"
        android:textAlignment="center"
        android:textSize="28dp"
```

```
        android:textStyle="bold"
        android:layout_marginBottom="10dp"
<Button
        android:id="@+id/b10"
        android:layout_width="105dp"
        android:layout_height="75dp"
        android:layout_marginLeft="5dp"
        android:layout_row="4"
        android:layout_column="1"
        android:gravity="center"
        android:textAlignment="center"
        android:textSize="28dp"
        android:textStyle="bold"
        android:layout_marginBottom="10dp"
android:background="@drawable/button_back
ground"android:text="."/>

    <Button
        android:id="@+id/b11"
        android:layout_width="105dp"
        android:layout_height="75dp"
        android:layout_marginLeft="5dp"
        android:layout_row="4"
        android:layout_column="2"
        android:gravity="center"
        android:textAlignment="center"
        android:textSize="28dp"
        android:textStyle="bold"
        android:layout_marginBottom="10dp"
android:background="@drawable/button_back
ground" android:text="bp"/>

    <Button
        android:id="@+id/b15"
        android:layout_width="105dp"
        android:layout_height="75dp"
        android:layout_marginLeft="5dp"
        android:layout_row="4"
        android:layout_column="3"
        android:gravity="center"
        android:textAlignment="center"
        android:textSize="28dp"
        android:textStyle="bold"
        android:layout_marginBottom="10dp"

android:background="@drawable/op_bg"
        android:textColor="#ffffff"
        android:text="/"/>
          <Button
        android:id="@+id/bac"
```

```
        android:layout_width="215dp"
        android:layout_height="75dp"
        android:layout_row="5"
        android:layout_column="0"
        android:layout_columnSpan="2"
        android:layout_marginLeft="5dp"
        android:text="AC"
        android:textColor="#ffffff”
android:background="@drawable/op_bg"
        android:gravity="center"
        android:textAlignment="center"
        android:textSize="28dp"
        android:textStyle="bold"
android:layout_marginBottom="10dp"/>

      <Button
        android:id="@+id/b16"
        android:layout_width="215dp"
        android:layout_height="75dp"
        android:layout_marginLeft="5dp"
        android:layout_row="5"
        android:layout_column="2"
        android:layout_columnSpan="2"
        android:gravity="center"
        android:textAlignment="center"
        android:textSize="28dp"
        android:textStyle="bold"
        android:layout_marginBottom="10dp"
 android:background="@drawable/op_bg"
        android:textColor="#ffffff"
        android:text="="/>
   </GridLayout>
</LinearLayout>
```

Result: The program has executed successfully and output is verified

OUTPUT



Date : 25/07/26

EXPERIMENT 4

Aim: To implement basic arithmetic operations (Addition, Subtraction, Multiplication and Division)

ALGORITHM

1. Start.
2. Create the user interface with two input fields and four buttons.
3. Enter the first and second numbers.
4. Select the required arithmetic operation.
5. Perform the selected operation:
6. Addition: \(R = A + B\)
7. Subtraction: \(R = A - B\)
8. Multiplication: \(R = A \times B\)
9. Division: \(R = A / B\)
10. If the second number is zero during division, display an error message.
11. Display the result.
12. Stop

MainActivity.java

```
package com.example.calculator;

import android.os.Bundle;

import android.view.View;

import android.widget.Button;

import android.widget.EditText;

import android.widget.TextView;

import androidx.appcompat.app.AppCompatActivity;

public class MainActivity extends AppCompatActivity {

   private EditText n1, n2;

   private TextView result;
```

```
@Override
protected void onCreate(Bundle savedInstanceState) {
    super.onCreate(savedInstanceState);
    setContentView(R.layout.activity_main);
    n1 = findViewById(R.id.n1);
    n2 = findViewById(R.id.n2);
    result = findViewById(R.id.result);
    Button add = findViewById(R.id.add);
    Button sub = findViewById(R.id.sub);
    Button mul = findViewById(R.id.mul);
    Button div = findViewById(R.id.div);
    View.OnClickListener calculate = v -> {
        String a = n1.getText().toString().trim();
        String b = n2.getText().toString().trim();
        if (a.isEmpty() || b.isEmpty()) {
            result.setText("Enter both numbers");
            return;
        }
        try {
            double x = Double.parseDouble(a);
            double y = Double.parseDouble(b);
            if (Double.isNaN(x) || Double.isInfinite(x)
                    || Double.isNaN(y) || Double.isInfinite(y)) {
                result.setText("Enter valid numbers");
                return;
            }
            double r;
```

```
        int id = v.getId();
        if (id == R.id.add) {
            r = x + y;
        } else if (id == R.id.sub) {
            r = x - y;
        } else if (id == R.id.mul) {
            r = x * y;
        } else if (id == R.id.div) {
            if (y == 0) {
                result.setText("Cannot divide by zero");
                return;
            }
            r = x / y;
        } else {
            return;
        }
        if (Double.isNaN(r) || Double.isInfinite(r)) {
            result.setText("Result out of range");
        } else {
            result.setText("Result: " + r);
        }
    } catch (NumberFormatException e) {
        result.setText("Enter valid numbers");
    }
};
add.setOnClickListener(calculate);
sub.setOnClickListener(calculate);
```

```
        mul.setOnClickListener(calculate);
        div.setOnClickListener(calculate);
    }
}
```

Activity_Main.Xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:gravity="center"
    android:padding="20dp">

    <EditText
        android:id="@+id/n1"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:minHeight="48dp"
        android:hint="Enter first number"
        android:inputType="numberDecimal|numberSigned"
        android:layout_marginBottom="12dp" />

    <EditText
        android:id="@+id/n2"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:minHeight="48dp"
```

```
    android:hint="Enter second number"
    android:inputType="numberDecimal|numberSigned"
    android:layout_marginBottom="20dp" />

<Button
    android:id="@+id/add"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:minHeight="48dp"
    android:text="ADD" />

<Button
    android:id="@+id/sub"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:minHeight="48dp"
    android:text="SUBTRACT" />

<Button
    android:id="@+id/mul"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:minHeight="48dp"
    android:text="MULTIPLY" />

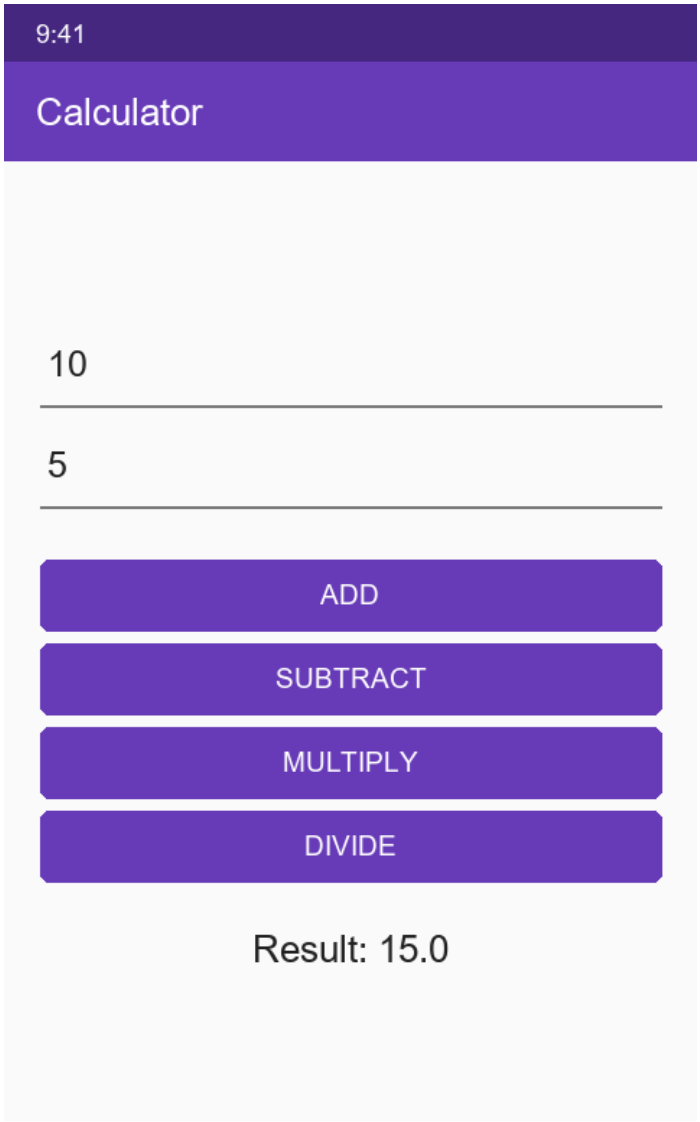
<Button
    android:id="@+id/div"
```

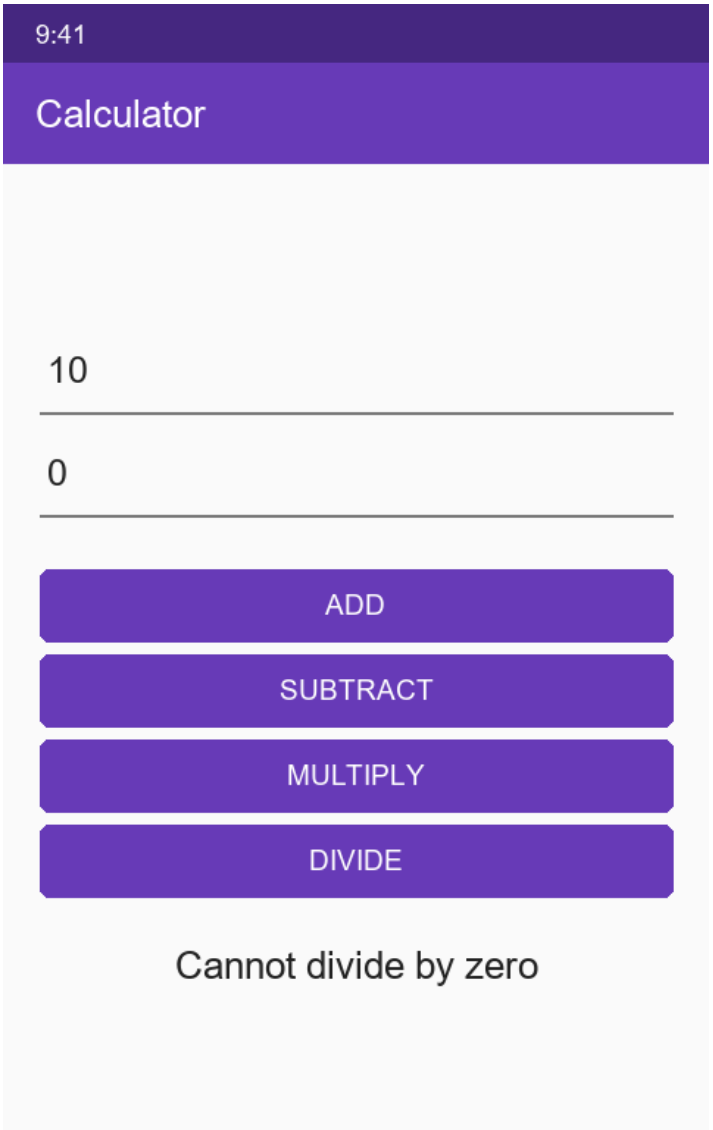
```
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:minHeight="48dp"
        android:text="DIVIDE" />

    <TextView
        android:id="@+id/result"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:minHeight="48dp"
        android:layout_marginTop="16dp"
        android:gravity="center"
        android:text="Result: "
        android:textSize="22sp"
        android:accessibilityLiveRegion="polite"  />
</LinearLayout>
```

Result: The program has executed successfully and output is verified

OUTPUT


9:41
Calculator
10
5
ADD
SUBTRACT
MULTIPLY
DIVIDE
Result: 15.0


9:41
Calculator
10
0
ADD
SUBTRACT
MULTIPLY
DIVIDE
Cannot divide by zero

Date : 15/08/26

EXPERIMENT 5

Aim : Create an app to demonstrate the usage of shared preferences by writing and reading data.

ALGORITHM

1.Start the application and display the Write Data button, Read Data button, and status TextView.

2. Initialize the controls and define the sample username and email.

3.Open the SharedPreferences file `sp_abc` in private mode.

4.When Write Data is clicked, obtain a SharedPreferences.Editor.

5. Store the username and email using their respective keys and call `apply()`.

6. Display "Written Data" in the status TextView.

7. When Read Data is clicked, retrieve the saved username and email using `getString()`.

8. Display the retrieved username and email in the status TextView.

9. Stop the application.

Program Code :

MainActivity.java

```
package com.example.sharedpref;

import android.os.Bundle;
import android.content.Intent;
import android.content.SharedPreferences;
import android.widget.*;
import androidx.appcompat.app.AppCompatActivity;

public class MainActivity extends AppCompatActivity {

   EditText name;
   Button save;

   @Override
   protected void onCreate(Bundle b) {
      super.onCreate(b);
      setContentView(R.layout.activity_main);
```

```
        name = findViewById(R.id.name);
        save = findViewById(R.id.save);

        save.setOnClickListener(v -> {
            SharedPreferences sp =
                getSharedPreferences("MyData", MODE_PRIVATE);

            sp.edit().putString("name",
                name.getText().toString()).apply();

            Intent i = new Intent(this, SecondActivity.class);
            startActivity(i);
        });
    }
}
```

Activity_home_page.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout
    xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:padding="20dp">

    <EditText
        android:id="@+id/name"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:hint="Enter Name"/>

    <Button
        android:id="@+id/save"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:text="Save and Next"/>

</LinearLayout>
```

SecondActivity.java

```
package com.example.sharedpref;

import android.os.Bundle;
import android.content.SharedPreferences;
import android.widget.*;
import androidx.appcompat.app.AppCompatActivity;

public class SecondActivity extends AppCompatActivity {

    @Override
```

```
    protected void onCreate(Bundle b) {
        super.onCreate(b);
        setContentView(R.layout.activity_second);

        TextView result = findViewById(R.id.result);

        SharedPreferences sp =
            getSharedPreferences("MyData", MODE_PRIVATE);

        String name = sp.getString("name", "No Data");

        result.setText("Welcome " + name);
    }
}
```

Activity_Second.Xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout
    xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:padding="20dp">

    <TextView
        android:id="@+id/result"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:textSize="25sp"/>

</LinearLayout>
```

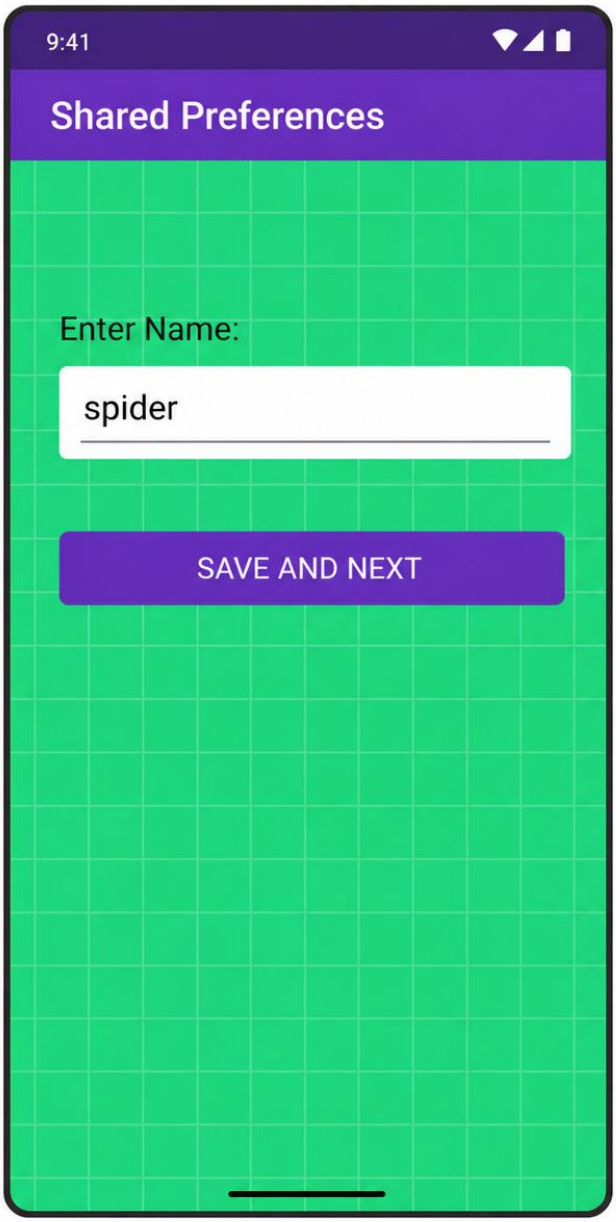
AndroidManifest.xml

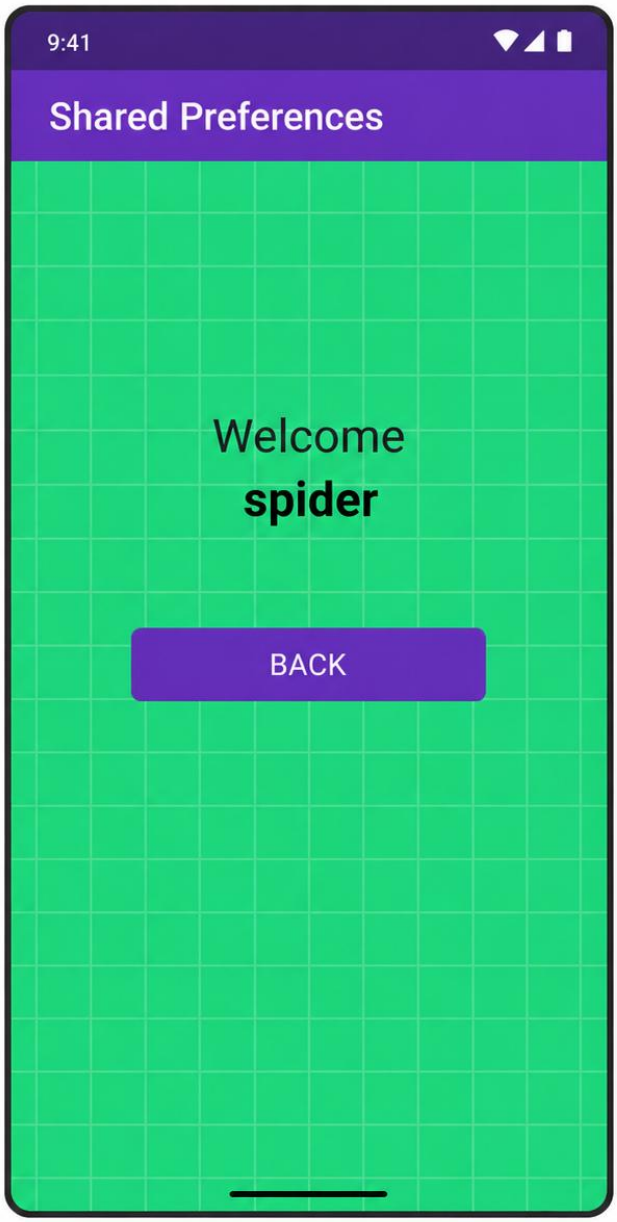
Add the second activity inside the <application> tag:

```
<activity
    android:name=".SecondActivity"
    android:exported="false" />
```

Result: The program has executed successfully and output is verified

OUTPUT


9:41
Shared Preferences
Enter Name:
spider
SAVE AND NEXT


9:41
Shared Preferences
Welcome
spider
BACK

Date : 17/08/26

EXPERIMENT 6

Aim : Create an app to demonstrate the usage of Checkbox and Radio buttons

ALGORITHM

1 Start the Android Studio project and create MainActivity.java.

2 Design the user interface in activity_main.xml with RadioButtons for gender selection, CheckBoxes for language selection, a Submit button, and a TextView to display the selected languages.

3 Declare and initialize the Button, RadioGroup, RadioButtons, CheckBoxes, and TextView using findViewById().

4 Set an OnClickListener for the Submit button.

5 Check which language CheckBoxes are selected and display the selected languages in the TextView.

6 Set an OnCheckedChangeListener for the RadioGroup to detect the selected gender.

7 Display the selected gender using a Toast message.

8 Run the application and verify the output.

Program Code :

MainActivity.java:

```
package com.example.radiocheck;
import androidx.appcompat.app.AppCompatActivity;
import android.os.Bundle;
import android.view.View;
import android.widget.Button;
import android.widget.CheckBox;
import android.widget.RadioButton;
import android.widget.RadioGroup;
import android.widget.TextView;
import android.widget.Toast;
public class MainActivity extends AppCompatActivity
{
Button b1;
RadioGroup rg;
RadioButton r1,r2;
CheckBox e,m,h;
TextView t;
@Override
protected void onCreate(Bundle savedInstanceState)
{
super.onCreate(savedInstanceState);
setContentView(R.layout.activity_main);
```

```
b1=(Button) findViewById(R.id.submit);
r1=(RadioButton) findViewById(R.id.male);
r2=(RadioButton) findViewById(R.id.female);
e=(CheckBox)findViewById(R.id.e);
m=(CheckBox)findViewById(R.id.m);
h=(CheckBox)findViewById(R.id.h);
t=(TextView)findViewById(R.id.tv);
b1.setOnClickListener(new View.OnClickListener()
{
@Override
public void onClick(View view)
{
String s[]=new String[3];
t.setText("Languages Known: ");
if(e.isChecked())
t.setText(t.getText().toString()+" "+e.getText());
if(m.isChecked())
t.setText(t.getText().toString()+" "+m.getText());
if(h.isChecked())
t.setText(t.getText().toString()+" "+h.getText());
}
});
rg=(RadioGroup) findViewById(R.id.rg);
rg.setOnCheckedChangeListener(new RadioGroup.OnCheckedChangeListener()
{
@Override
public void onCheckedChanged(RadioGroup group, int checkedId)
{
if(checkedId==R.id.female)
{ Toast.makeText(getApplicationContext(),r2.getText().toString(),Toast.LENGTH_SHORT).show();
}
else
{ Toast.makeText(getApplicationContext(),r1.getText().toString(),Toast.LENGTH_SHORT).show();
}
}
});
}
}
```

Activity_home_page.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout
xmlns:android="http://schemas.android.com/ap
k/res/android"
xmlns:app="http://schemas.android.com/apk/re
s-auto"
xmlns:tools="http://schemas.android.com/tool"
 android:layout_width="match_parent"
 android:layout_height="match_parent"
```

```
 tools:context=".MainActivity">
 <RelativeLayout
 android:layout_width="match_parent"
 android:layout_height="match_parent">
 <TextView
 android:layout_width="wrap_content"
 android:layout_height="30dp"
 android:layout_marginTop="50dp"
 android:layout_marginLeft="150dp"
 android:id="@+id/h1"
 android:text="Registration"
 android:textSize="24dp"/>
 <TextView
 android:layout_width="wrap_content"
 android:layout_height="30dp"
 android:layout_marginTop="100dp"
 android:layout_marginLeft="50dp"
android:id="@+id/gender"
 android:text="Choose Gender"
 android:textSize="20dp"/>
 <RadioGroup
 android:layout_width="250dp"
 android:layout_height="wrap_content"
 android:id="@+id/rg"
 android:layout_marginTop="150dp"
 android:layout_marginLeft="50dp">
 <RadioButton
 android:layout_width=
"wrap_content"
 android:layout_height=
"wrap_content"
 android:text="Male"
 android:id="@+id/male"/>
 <RadioButton
 android:layout_width=
"wrap_content"
 android:layout_height=
"wrap_content"
android:text="Female"
 android:id="@+id/female"/>
 </RadioGroup>
 <TextView
 android:layout_width="wrap_content"
android:layout_height="30dp"
 android:layout_marginTop="275dp"
 android:layout_marginLeft="50dp"
 android:id="@+id/lang"
 android:text="Select Known Languages"
 android:textSize="20dp"/>
 <CheckBox
```

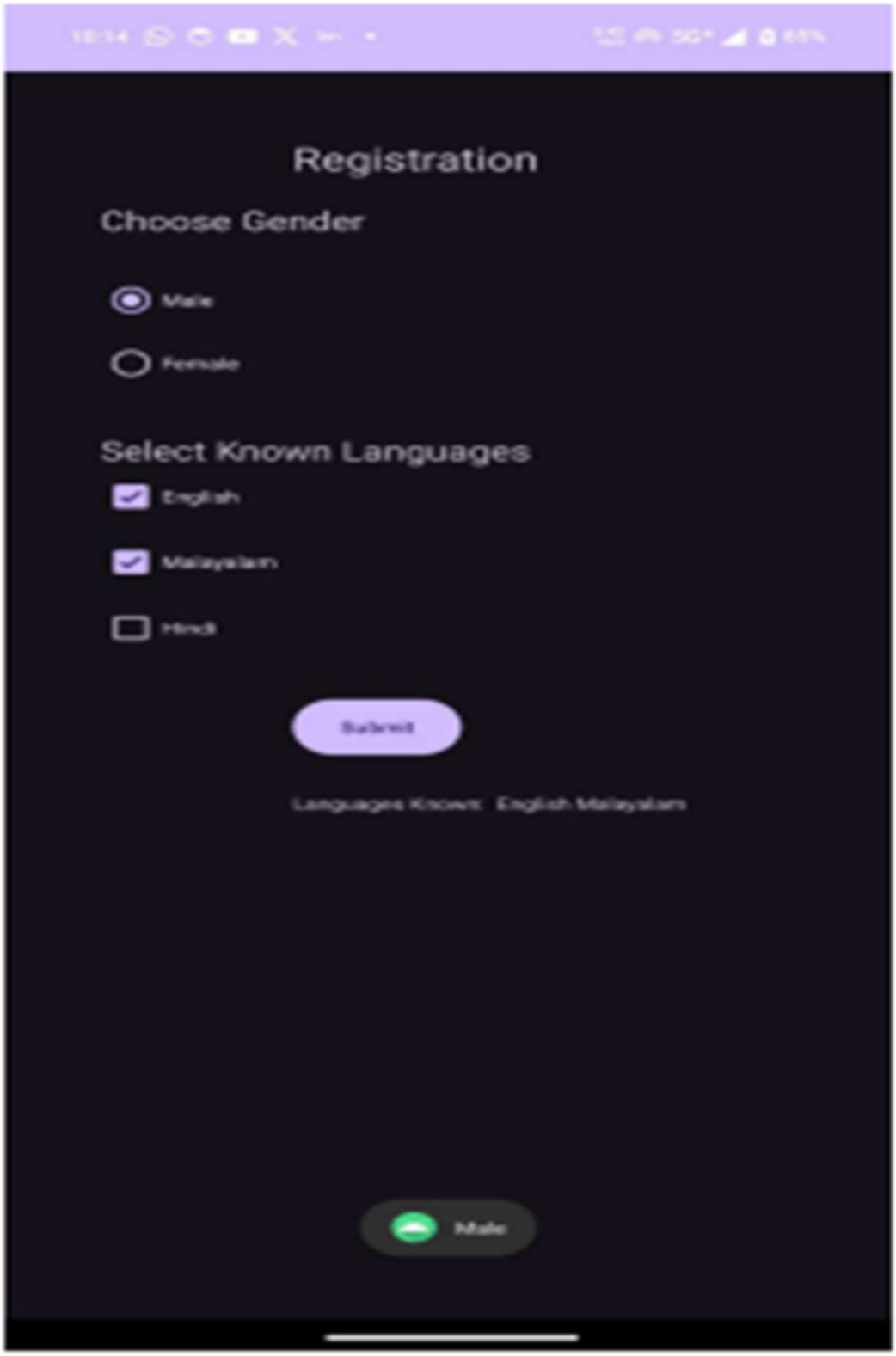
```
 android:id="@+id/e"
 android:layout_width="wrap_content"
 android:layout_height="wrap_content"
 android:text="English"
 android:layout_marginTop="300dp"
 android:layout_marginLeft="50dp"/>
 <CheckBox
 android:id="@+id/m"
 android:layout_width="wrap_content"
 android:layout_height="wrap_content"
 android:text="Malayalam"
 android:layout_marginTop="350dp"
 android:layout_marginLeft="50dp"/>
 <CheckBox
 android:id="@+id/h"
android:layout_width="wrap_content"
 android:layout_height="wrap_content"
 android:text="Hindi"
 android:layout_marginTop="400dp"
 android:layout_marginLeft="50dp"/>
 <Button
 android:layout_width="wrap_content"
 android:layout_height="50dp"
 android:layout_marginTop="475dp"
 android:layout_marginLeft="150dp"
 android:text="Submit"
 android:id="@+id/submit"/>
```

Result: The program has executed successfully and output is verified.

```
 <TextView
 android:layout_width="wrap_content"
 android:layout_height="50dp"
 android:layout_marginTop="550dp"
 android:layout_marginLeft="150dp"
 android:text=""
 android:id="@+id/tv"/>
 </RelativeLayout>
</LinearLayout>
```

Result: The program has executed successfully and output is verified

OUTPUT


Registration
Choose Gender
Male
Female
Select Known Languages
English
Malayalam
Hindi
Submit
Languages Known: English Malayalam
Male

Date : 19/08/26

EXPERIMENT 7

Aim : Create an app to demonstrate the usage of implicit intent by opening a website form the app

ALGORITHM

Implicit Intent Algorithm
1. Start the application and display the layout containing the Open College Website button.
2. Initialize the button and status TextView.
3. Wait for the user to tap the button.
4. Create an implicit Intent with the action Intent.ACTION_VIEW.
5. Set the Intent data to the URI https://lmcst.ac.in.
6. Call startActivity with the Intent so a suitable browser opens the college website.
7. Update the status TextView and allow the user to return to the application. Implicit Intent

Program Code :

MainActivity.java:

```
import android.content.Intent;

import android.net.Uri;

import android.os.Bundle;

import android.view.View;

import android.widget.Button;

import android.widget.TextView;

import androidx.activity.EdgeToEdge;

import androidx.appcompat.app.AppCompatActivity;
```

```
public class MainActivity extends AppCompatActivity

{

Button b;

TextView t;

@Override

protected void onCreate(Bundle savedInstanceState)

{

super.onCreate(savedInstanceState);

EdgeToEdge.enable(this);

setContentView(R.layout.activity_main);

b=(Button) findViewById(R.id.b1);

t=(TextView) findViewById(R.id.t1);

b.setOnClickListener(new View.OnClickListener()

{

@Override

public void onClick(View view) {

Intent intent = new Intent(Intent.ACTION_VIEW);

intent.setData(Uri.parse("https://lmcst.ac.in  "));

startActivity(intent);
```

```
t.setText("Youtube Opened");

}

});

}

}
```

Activity_home_page.xml

```
<?xml version="1.0" encoding="utf-8"?>

<LinearLayout

xmlns:android="http://schemas.android.com/ap

k/res/android"

xmlns:app="http://schemas.android.com/apk/re

s-auto"

xmlns:tools="http://schemas.android.com/tools

" android:id="@+id/main"

android:layout_width="match_parent"

android:layout_height="match_parent"

tools:context=".MainActivity">

<RelativeLayout

android:layout_width="wrap_content"
```

```
android:layout_height="wrap_content">

<TextView

android:layout_width="wrap_content"

android:layout_height="wrap_content"

android:text="IMPLICIT Intent \nTo

Open College Website"

android:layout_marginLeft="115dp"

android:layout_marginTop="100dp"

android:textSize="20sp">

</TextView>

<Button

android:id="@+id/b1"

android:layout_width="wrap_content"

android:layout_height="wrap_content"

android:layout_marginLeft="95dp"

android:layout_marginTop="200dp"

android:text="Open College Website">

</Button>

<TextView
```

```
android:id="@+id/t1"

android:layout_width="wrap_content"

android:layout_height="wrap_content"

android:text=""

android:layout_marginTop="300dp"

android:layout_marginLeft="150dp"

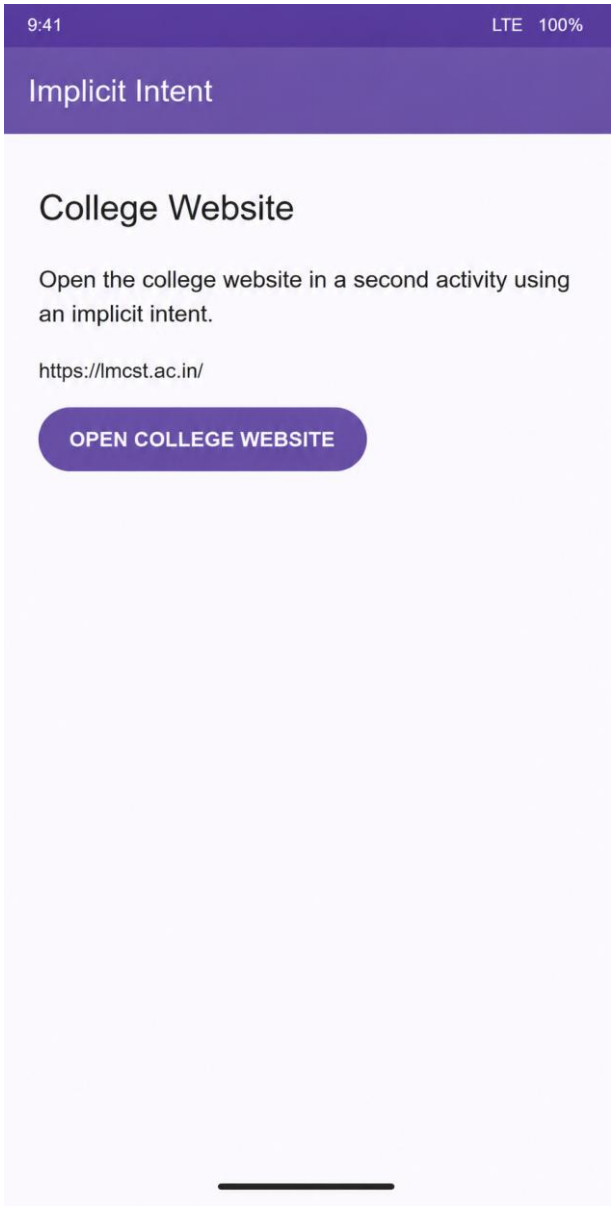
android:textSize="20sp"></TextView>

</RelativeLayout>

</LinearLayout>
```

Result: The program has executed successfully and output is verified

OUTPUT


9:41
LTE 100%
Implicit Intent
College Website
Open the college website in a second activity using an implicit intent.
https://lmcst.ac.in/
OPEN COLLEGE WEBSITE


9:41
LTE 100%
College Website
https://lmcst.ac.in/
8921040851
principal@lmcst.ac.in
For Admission and Queries: 8921040851
Placement Coordinator: 9633738043
Office: 6282098290
LOURDES MATHA
College Code: LMC
द्रव नोदन प्रणाली केंद्र
UID PROPULSION SYSTEMS CENTRE
Announcements
Latest Updates
ADD ON COURSE

Date : 04/09/26

EXPERIMENT 8

AIM:To develop an Android application using ArrayAdapter with ListView to display a list of vehicles and show a Toast message when an item is selected.

ALGORITHM

1. Start the Android project and create MainActivity.java.
2. Design the UI using activity_main.xml with a ListView.
3. Declare a string array containing Bike, Scooter, Auto, Bus and Cycle.
4. Find the ListView using findViewById().
5. Create an ArrayAdapter using the context, predefined layout and string array.
6. Set the adapter to the ListView using setAdapter().
7. Set an OnItemClickListener to detect the selected item.
8. Display a Toast message using the showToast() method.
9. Run the application and verify the list and Toast messages.
10. Stop..

Program Code

MainActivity.java

```
package com.example.adaptor;

import android.os.Bundle;
import android.view.View;
import android.widget.*;
import androidx.appcompat.app.AppCompatActivity;

public class MainActivity extends AppCompatActivity {

   @Override
   protected void onCreate(Bundle savedInstanceState) {
      super.onCreate(savedInstanceState);
      setContentView(R.layout.activity_main);

      String[] data = {"Bike", "Scooter", "Auto", "Bus", "Cycle"};

      ListView list = findViewById(R.id.listview);

      ArrayAdapter<String> adapter = new ArrayAdapter<>(
            this,
```

```
            android.R.layout.simple_list_item_1,
            data
        );

        list.setAdapter(adapter);

        list.setOnItemClickListener((parent, view, position, id) -> {
            showToast("You selected: " + data[position]);
        });
    }

    private void showToast(String message) {
        Toast.makeText(this, message, Toast.LENGTH_SHORT).show();
    }
}
```

Activity_Main.Xml

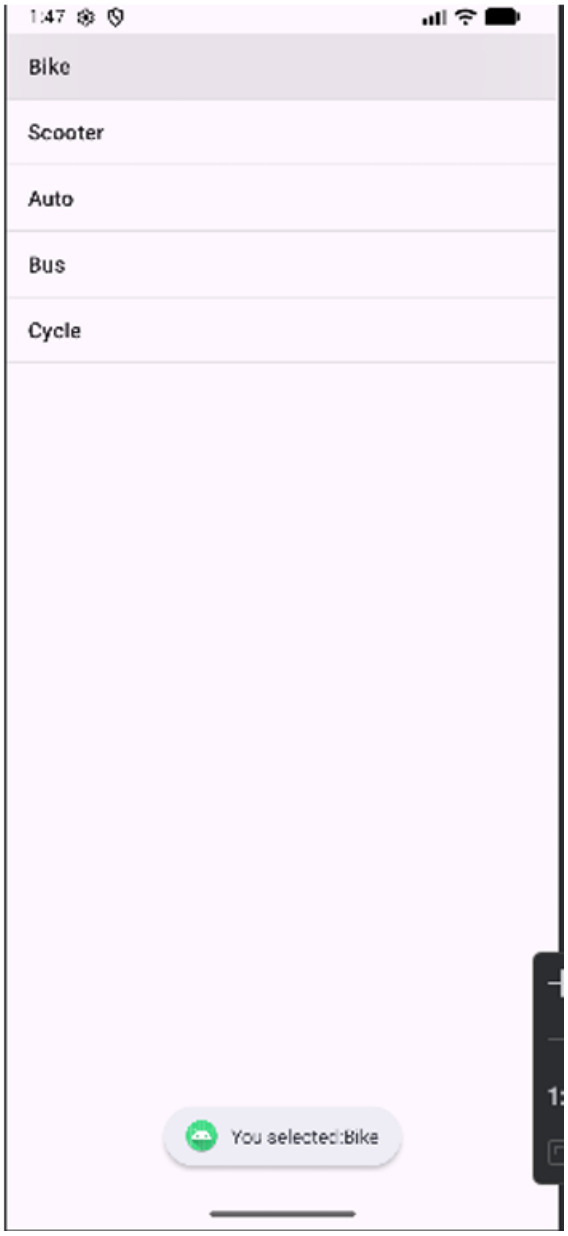
```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout
    xmlns:android="http://schemas.android.com/apk/res/android"
    android:id="@+id/main"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical">

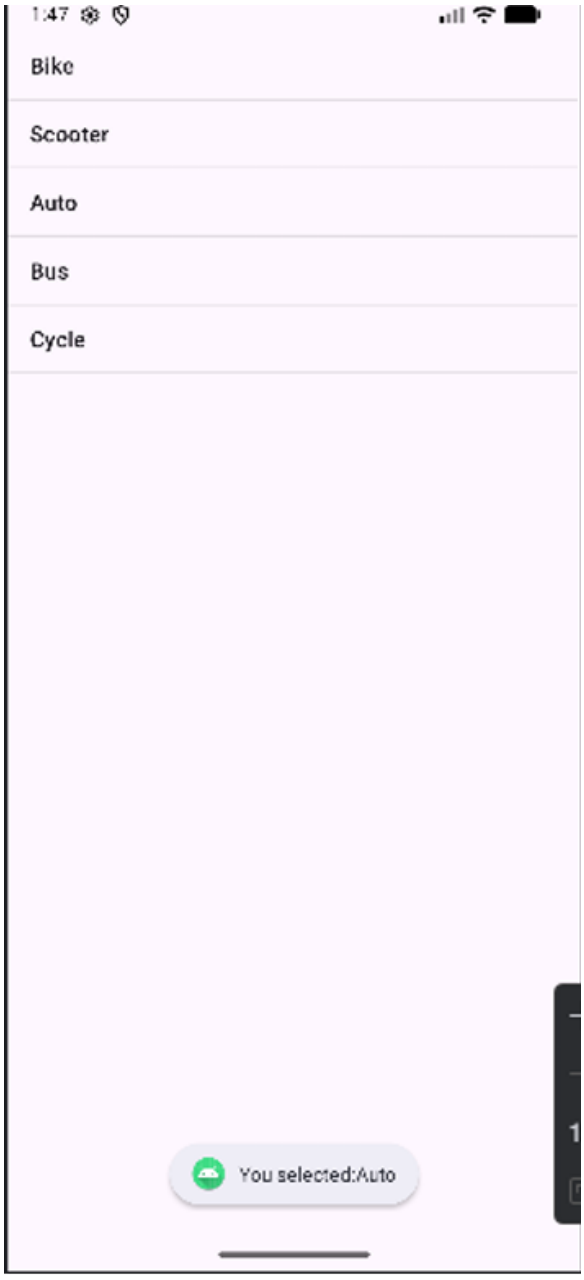
    <ListView
        android:id="@+id/listview"
        android:layout_width="match_parent"
        android:layout_height="match_parent" />

</LinearLayout>
```

Result: The program has executed successfully and output is verified

OUTPUT


1:47
Bike
Scooter
Auto
Bus
Cycle
You selected:Bike


1:47
Bike
Scooter
Auto
Bus
Cycle
You selected:Auto

Date : 16/09/26

EXPERIMENT 9

Aim : Create an app to create a spinner and select an item from it.

Algorithm

1. Start the application and display the Spinner layout.

2. Initialize the Spinner and the output TextView.

3. Create a list containing mango, apple, pineapple and banana.

4. Create an ArrayAdapter using the fruit list and a spinner dropdown layout.

5. Assign the adapter to the Spinner.

6. Register an item selection listener.

7. When a fruit is selected, obtain it using the selected position and display Selected Fruit followed by its name.

8. If no item is selected, display the selection prompt; continue accepting selections until the application is closed.

Program Code :

MainActivity.java:

```
package com.example.spinner;
import androidx.appcompat.app.AppCompatActivity;
import android.os.Bundle;
import android.view.View;
import android.widget.AdapterView;
import android.widget.ArrayAdapter;
import android.widget.Spinner;
import android.widget.TextView;
import android.widget.Toast;
import java.util.ArrayList;
public class MainActivity extends AppCompatActivity
{
Spinner s;
TextView t1;
@Override
protected void onCreate(Bundle savedInstanceState)
{
super.onCreate(savedInstanceState);
setContentView(R.layout.activity_main);
s=(Spinner)findViewById(R.id.sp);
t1=(TextView) findViewById(R.id.t1);
ArrayList <String> al=new ArrayList<String>();
al.add("mango");
al.add("apple");
```

```
al.add("pineapple");
al.add("banana");
ArrayAdapter ad=new ArrayAdapter<>(this, androidx.appcompat.R.layout.support_simple_
spinner_dropdown_item,al);
s.setAdapter(ad);
s.setOnItemSelectedListener(new AdapterView.OnItemSelectedListener()
{
@Override
public void onItemSelected(AdapterView<?> parent, View view, int position, long id)
{
t1.setText("Selected Fruit: "+al.get(position));
}
@Override
public void onNothingSelected(AdapterView<?> parent)
{
t1.setText("Select a Fruit !!!!");
}
});
}
}
```

Activity_home_page.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout
xmlns:android="http://schemas.android.com/ap
k/res/android"
xmlns:app="http://schemas.android.com/apk/re
s-auto"
xmlns:tools="http://schemas.android.com/tools
"
android:id="@+id/main"
android:layout_width="match_parent"
android:layout_height="match_parent"
tools:context=".MainActivity">
<RelativeLayout
android:layout_width="wrap_content"
android:layout_height="wrap_content">
<TextView
android:layout_width="wrap_content"
android:layout_height="wrap_content"
android:layout_marginLeft="150dp"
android:layout_marginTop="50dp"
android:text="Spinner"
android:textSize="42dp"></TextView>
<TextView
android:layout_width="wrap_content"
android:layout_height="wrap_content"
android:layout_marginLeft="85dp"
android:layout_marginTop="135dp"
```

```
android:text="Select Fruit:"
android:textSize="20dp">
</TextView>
```

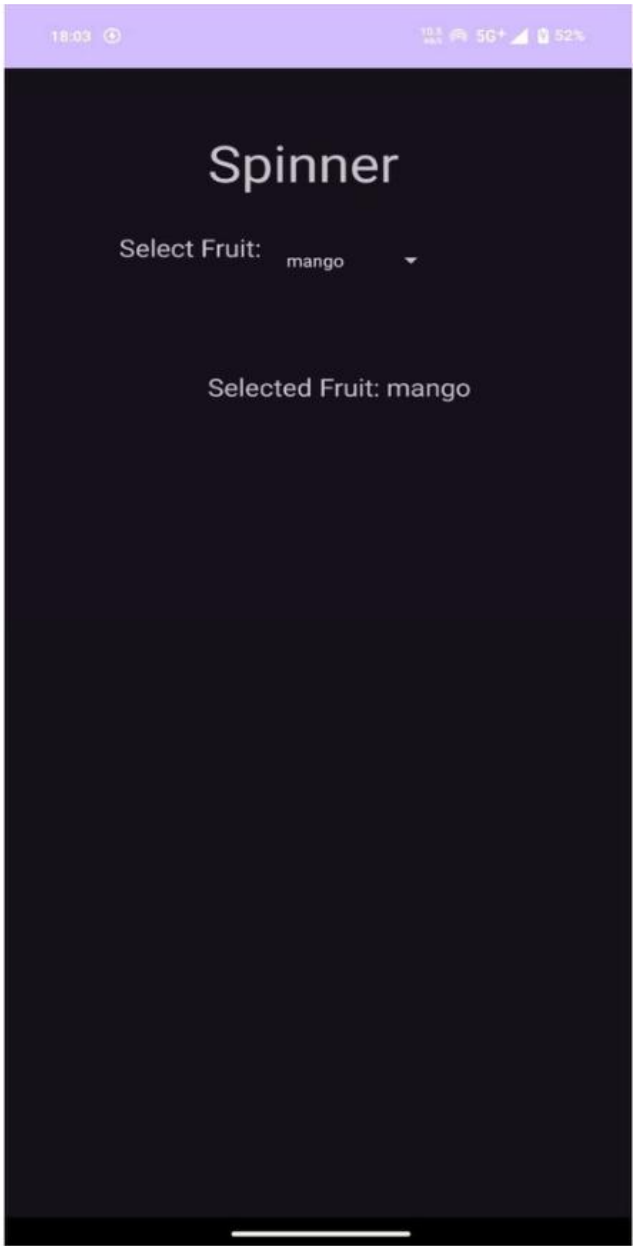
Result: The program has executed successfully and output is verified.

```
<Spinner
android:layout_width="wrap_content"
android:layout_height="wrap_content"
android:layout_marginLeft="200dp"
android:layout_marginTop="135dp"
android:id="@+id/sp"></Spinner>
<TextView
android:id="@+id/t1"
android:layout_width="wrap_content"
android:layout_height="wrap_content"
android:layout_marginLeft="150dp"
android:layout_marginTop="250dp"
android:text=""
android:textSize="20dp"></TextView>
</RelativeLayout>
</LinearLayout>
```

Result: The program has executed successfully and output is verified

OUTPUT


18:03
5G+
52%
Spinner
Select Fruit:
mango
apple
pineapple
banana
Selected Fruit: mango


18:03
5G+
52%
Spinner
Select Fruit:
mango
Selected Fruit: mango